# Can we trust the GIPL 2.0 permafrost projections?

GIPL 2.0 ground temperature projections for Alaska (SNAP, University of Alaska Fairbanks): 1 km, 2021–2120, three climate inputs (5-model average, GFDL-CM3, NCAR-CCSM4), RCP 4.5 and 8.5.

**Short answer: not as delivered.** We compared GIPL 2.0 with a satellite-based product (ESA CCI Permafrost), 90 boreholes and about 3,000 CUSP field sites, and checked how the model behaves over time. We found four problems:

1 **Stuck cells.** Across 46,000 km², mostly in the Interior, the ground from 1 m down never warms in 100 years while the surface does.
2 **Too cold.** About 1–2 °C colder than every independent reference, by different amounts in different places.
3 **Start-up and duplicated runs.** The first years jump about 2 °C, and through 2035, runs that should differ do not.
4 **Out of step with the air.** GIPL's warming does not follow the air temperature projections for the same scenarios.

These problems are in every model and scenario. The run is undocumented, so they can be neither explained nor corrected.

## 1 Stuck cells: ground that never warms

In 1,172 patches covering **46,000 km² (4.6 % of Alaska's permafrost area)**, GIPL's temperature at 1 m changes by less than 0.1 °C from 2021 to 2120, under every climate model and scenario. **88 % of this area is in the Interior, where it covers 10 % of the permafrost area, or 19 % with the 2 km border.** The ground temperature is also stuck at every deeper level we checked (2, 3 and 5 m), and 0.5 m warms only 0.4 °C over the century (about 5 °C elsewhere). Each cell holds its own fixed value (−5 to −1 °C at 1 m). Yet their surface warms *faster* than nearby cells' (1.5–3.5 °C warmer by 2081–2100), and cells within 2 km are partly affected. Satellite data show nothing unusual there, and soil and land-cover maps do not explain the pattern.

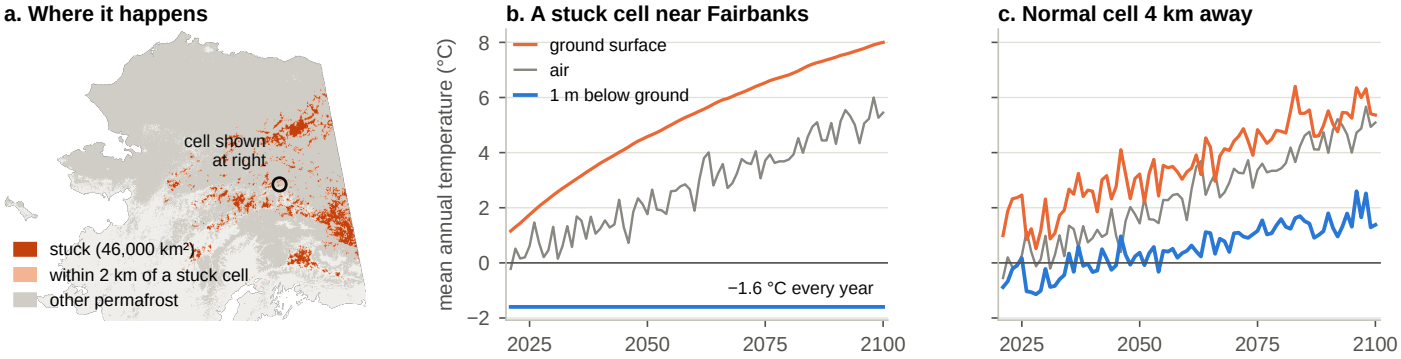



**Figure 1.** (a) Cells where 1 m temperature changes by < 0.1 °C, 2021–2120. (b, c) A stuck and a normal cell 4 km apart (5-model average, RCP 8.5).

## 2 Too cold, by different amounts in different places

Three independent checks agree:

- **Satellite-based product (ESA CCI):** allowing for about 1 °C of warming between the two products' decades, GIPL is 1.2 °C colder at the surface and 1.9 °C colder at 1 and 2 m (statewide median), and colder in 82–91 % of permafrost cells (Fig. 2a).
- **90 monitoring boreholes** (mostly 2015–2024): GIPL is 2.2, 2.1 and 1.6 °C too cold at the surface, 1 m and 2 m; ESA CCI is within 0.5 °C (Fig. 2b). Even GIPL's 2026–2035 values, a decade later, are colder than what the boreholes measured.
- **About 3,000 field sites** from the CUSP database, where crews recorded whether permafrost was present: at 1 m, GIPL shows frozen ground at 71 % of the 1,532 sites where none was found (ESA CCI: 19 %; Fig. 2c).

The error varies by about 3 °C from cell to cell and is largest (2–3 °C) near 0 °C (the frozen/thawed threshold). No single correction can remove it.

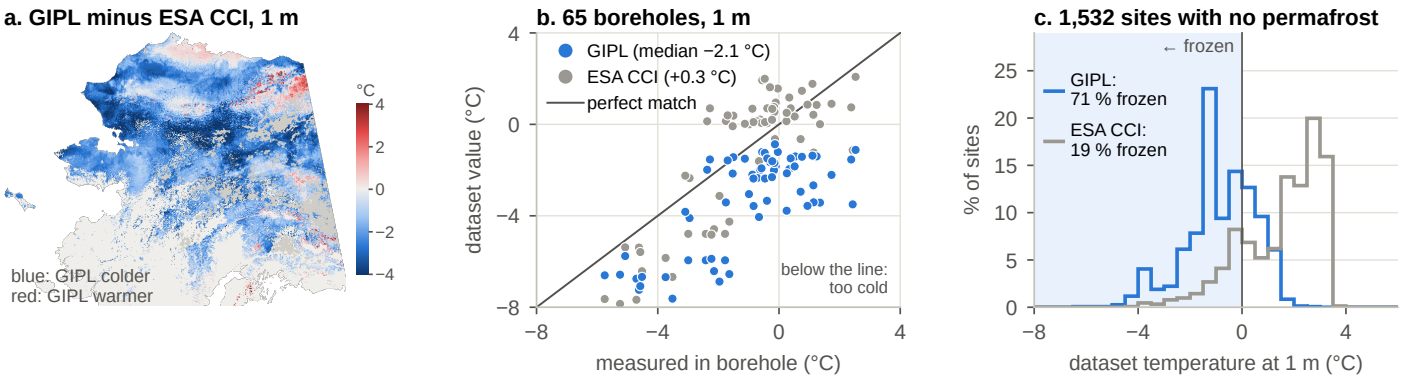



**Figure 2.** (a) GIPL 2026–2035 minus ESA CCI 2014–2023 at 1 m, after removing the projected warming in between (RCP 8.5). (b) Measured vs. dataset temperature, same years. (c) CUSP field sites where crews found no permafrost.

## 3 An unexplained start-up, and runs that should differ but do not

GIPL starts in 2021 far from where it settles. At 1 m the statewide median climbs 1.8 °C in two years (Fig. 3a); real ground does not warm that fast. The starting points are mixed up: the 5-model average's two scenarios start from different ground temperatures although their climates are still nearly the same in 2021 (Fig. 3a), yet four of the six runs (5-model average and GFDL-CM3 under RCP 4.5, NCAR-CCSM4 under both) start from identical values in every cell (Fig. 3b). In 2023 the NCAR-CCSM4 run's surface spikes to +3.6 °C (RCP 4.5) and +5.1 °C (RCP 8.5), then drops back (Fig. 3b). Its air temperature shows no such jump: under RCP 8.5, 2023 is the coldest year of 2021–2035. Then, **through 2035, runs that should differ share one sequence of warm and cold years:**

- GFDL-CM3 (one climate model) is **identical** to the 5-model average under RCP 4.5, in every cell and year (Fig. 3c).
- GFDL-CM3 under RCP 8.5 repeats nearly the same values one year earlier (typically within 0.05 °C; Fig. 3d).
- The 5-model average's two scenarios rise and fall together (correlation 0.97; Fig. 3a), although their air temperatures do not (−0.08).

The runs separate only from 2036. GIPL's 2021–2035 values are therefore not projections for a particular model or scenario, and comparing them with observations is unreliable.

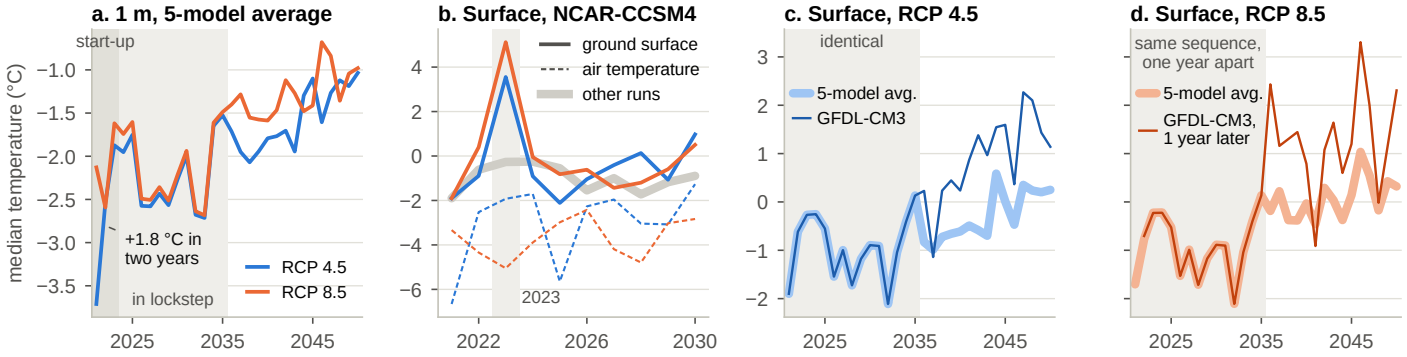



**Figure 3.** Statewide median of permafrost cells (stuck cells excluded). (a) Start-up at 1 m, then lockstep. (b) NCAR-CCSM4's surface (solid) spikes in 2023; its air temperature (dashed) does not. (c, d) Until 2035, GFDL-CM3 matches the 5-model average: exactly under RCP 4.5, one year apart under RCP 8.5.

## 4 Out of step with the air

GIPL converts air temperature and snow into ground temperature, so its warming should broadly follow the air warming of each scenario, damped or amplified by snow. It does not:

- From 2036 to 2070, GIPL's surface warms 0.5 °C per decade under *both* scenarios, while SNAP's air projections warm 0.3 (RCP 4.5) and 0.9 °C (RCP 8.5) per decade (Fig. 4).
- By 2081–2100, the extra 2.6 °C of air warming under RCP 8.5 shows up as only 1.5 °C of extra warming at the surface and 0.9 °C at 1 m.
- GIPL's year-to-year swings barely follow the air's (median correlation per cell between −0.45 and +0.49).

Snow explains part of this: the projected air warming is strongest in winter, under the snowpack. The rest may mean that GIPL was driven by different climate data than SNAP's published air temperature projections for the same models. Without documentation this cannot be checked.

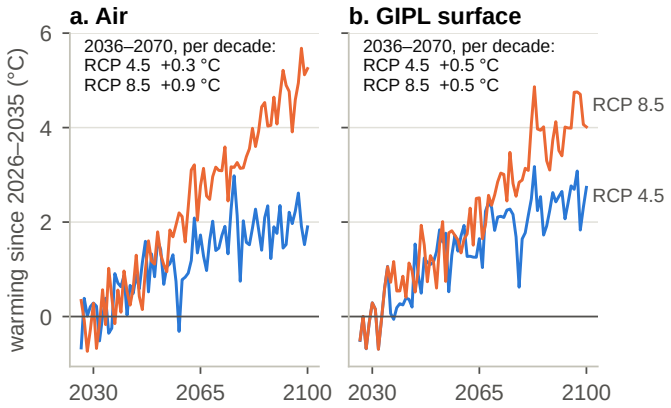



**Figure 4.** Warming since 2026–2035, statewide median of permafrost cells, 5-model average. (a) SNAP 2 km air temperature projections. (b) GIPL ground surface.

**What still holds.** GIPL's spatial pattern is sound: it ranks colder and warmer places as well as ESA CCI does when checked against the field sites. Its long-term warming is plausible in size. Two workarounds reduce the damage but do not remove it. Masking stuck cells and their 2 km border drops about 9 % of Alaska's permafrost area and cannot rule out milder effects elsewhere. Using only GIPL's change over time, added to an observation-based starting level, removes the cold bias but keeps problems 3 and 4.

**Bottom line.** GIPL 2.0 ground temperatures should not be used as absolute values (how cold the ground is, or whether it is frozen) anywhere in Alaska, and its projected changes only with the caveats above. The problems sit in the delivered data, and their causes cannot be identified: no documentation exists for this 1 km run (the only method reference describes an earlier, coarser run), including its climate inputs, starting conditions, and soil and snow settings.

**Sources and methods.** GIPL 2.0 (SNAP, University of Alaska Fairbanks): mean annual ground temperature at the surface, 0.5, 1 and 2 m, from SNAP's source GeoTIFFs (identical to its Rasdaman service). ESA CCI Permafrost v5.0, 1 km, 2003–2023. SNAP 2 km downscaled air temperature (CMIP5 5-model average and NCAR-CCSM4; CRU TS 4.08 for observed years). Boreholes: Thermal State of Permafrost in North America, UAF Permafrost Laboratory (Arctic Data Center); 90 Alaska sites with at least one full year of data, outside stuck cells and their border. Field sites: CUSP v1.2 (CommUnity near-Surface Permafrost), 3,012 GIPL cells with records from 2000 on and searches reaching at least 1 m. Stuck cell: annual 1 m temperature range over 2021–2120 below 0.1 °C in at least one run (97 % of the area is stuck in all six); 2, 3 and 5 m checked in the 5-model average, RCP 8.5. Regions: USDA general soil map physiography. Comparisons use permafrost cells (ESA CCI permafrost fraction > 0) outside stuck cells and their 2 km border (724,326 cells) and report medians. Because GIPL and ESA CCI describe different decades, the projected air warming between them (about +1 °C) is removed before comparing; the borehole result holds without this step. A borehole or field site is a point and a cell is 1 km², so single sites can differ from their cell by several degrees; medians over many sites are the meaningful numbers. ESA CCI is itself model-based and not error-free (about 0.5 °C on average against the boreholes). Ranking skill against field sites (area under the ROC curve): GIPL 0.81, ESA CCI 0.80–0.82. Detailed technical notes are available on request.